

Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Architected an LLM-based autonomous agent planning system (OpenAI Agents SDK, MCP tool-use) that decomposes goals into tool-use chains and executes them in production, handling millions of daily transactions at 99%+ uptime.
- Designed and tuned agentic workflows across multiple multi-modal LLMs (GPT, Claude, Llama, Gemini), selecting and tuning models per use case for production quality and cost.
- Built GPU/HPC cluster infrastructure in C++ and Go, scheduling jobs across Slurm and InfiniBand-networked nodes to support large-scale model training and inference.
- Designed system architecture for a pluggable distributed executor routing pipeline nodes across Docker and AWS Lambda, with an S3 artifact store eliminating redundant recomputation.
- Deployed a RAG pipeline on AWS SageMaker for real-time decision support at sub-200ms p99 latency, with an RTO under 2 minutes.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Built a model observability framework tracking drift, latency, and eval regressions in production, enabling governance-aligned rollback workflows.
- Improved retrieval recall 61% to 89% on a 2K-query eval set through systematic experimentation with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system with autonomous planning over tool-use chains; routes sub-tasks to specialized sub-agents and synthesizes findings into a single report.

LLaMA 2 Fine-Tuning Pipeline for Financial Documents

github.com/ganeshhgupta/llama2-finance-finetuning

- Post-trained an open-source LLM (LLaMA 2) with LoRA and PEFT for domain-specific document understanding, running systematic hyperparameter experiments.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Built a Raft-consensus distributed storage engine in C++ from scratch, implementing log compaction and linearizable reads under concurrent writes.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: C++, Python, Go, Java, TypeScript, SQL

GenAI: LLMs, Multi-Modal Models, LLM-Based Autonomous Agent Planning, Agentic AI, OpenAI Agents SDK, MCP, LoRA/PEFT

Systems: Data Structures & Algorithms, Software Design & Architecture, Distributed Systems, Testing

Cloud & DevOps: AWS (EC2, S3, SageMaker), GCP (Cloud Run), Docker, Kubernetes, CI/CD, PostgreSQL

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer